Multi-Turn Dialogue Response Selection via Utterance-oriented Strategy

Abstract

The Dialogue System Technology Challenge(DSTC) has high authority and popularity in the field of dialogue system. The outstanding performance of pre-trained models in dialogue systems has been proven, represented by Bidirectional Encoder Representation from Transformers(BERT). However, BERT can only focus on the semantic information between tokens during training, which makes it not able to fully understand the relationship between utterances. To address this issue, we propose an Utterance-oriented Strategy(UOS). On one hand, we use a more fine-grained approach in post-training stage, which can better capture utterance-to-utterance coherence and semantic information in dialogues. On the other hand, in fine-tuning stage, an utterance-level multiplexed matching network is designed, allowing the model to further capture the relationship at the utterance level, making full use of the multi-level information. To verify the effectiveness of the proposed UOS, we test it on four benchmark datasets. The experimental results show that our UOS achieves performance improvement.

1. Introduction

Natural language dialogue systems, also known as chatbots, are used in a wide range of domains. Natural language dialogue systems involve many typical tasks in the NLP domain, and a large number of researchers have been working on this task since NLP was formally proposed [11, 15, 16, 24]. In particular, most of the research in recent years has been based on deep learning method to model natural language dialogue systems, which has shown excellent performance [10, 13, 17].

A dialogue system can be viewed as a mapping from a user message U to a robot response R , i.e., $R = \phi(U)$, where $\phi(\cdot)$ is the mapping function. $\phi(\cdot)$ is often modeled using a Pipeline approach. The Pipeline approach gradually breaks down the dialogue understanding into a three-step chain structure of natural language understanding (NLU), dialogue management module (DM, including dialogue state tracking (DST) and policy learning (PL)), and natural language generation (NLG), and designs different models to solve the corresponding problems according to the characteristics of different modules. Among them, NLU [1, 3, 22] recognizes and parses the text information and converts it into a computer-understandable structured semantic representation; DST [8, 20] maintains and updates the current dialogue state based on the dialogue history and the semantic representation output by NLU module; PL [18, 25]performs the next system action based on the current dialogue state; NLG [2, 4] converts the semantic representation output by the dialogue policy into natural language text.

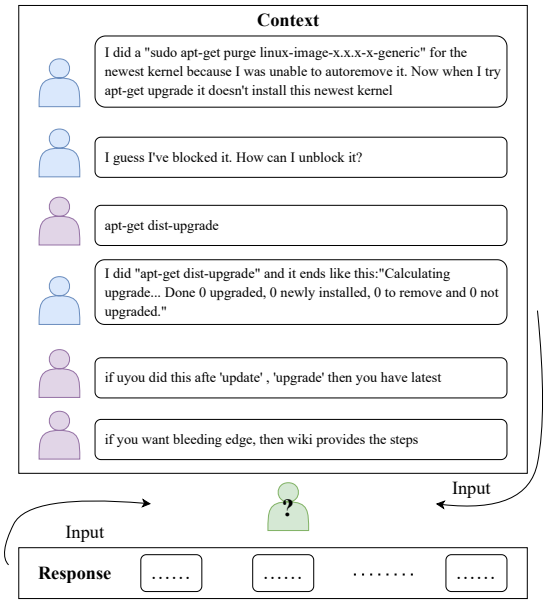

Figure 1. An example of multi-turn response selection.

The goal of our project is to propose a feasible solution for DSTC8 Track II (NOESIS II) [9]. As shown in the Fig. 1, NOESIS II is a typical dialogue system task that provides two specialized domain datasets, Ubuntu and Advising, for select-

ing the next round of responses in a multi-turn dialogue. Specifically, the input of a multi-turn dialogue is the dialogue context and 100 candidate responses, the output is the scoring ranking of 100 responses, and the evaluation metrics are mean-reverse ranking (MRR) and Top10 recall. NOESIS II differs from dialogue systems in that the NLU phase of the NOESIS II task requires understanding the semantic information of both the dialogue and the answers, while there are two main difficulties in NOESIS II, one is that the datasets are vertical domain questions and answers, and the model must have a certain understanding of domain expertise in order to answer correctly, for which the model needs to be well pre-trained in the domain; the other is that there are as many as 100 candidate responses for each conversation, which makes matching difficult. The second is that there are as many as 100 candidate responses in each conversation, which makes matching very difficult.

We have extensively investigated the sota model for multi-turn response selection, and we found that most existing studies use post train Bert to solve the multi-turn response selection task [6, 21]. These methods can understand the domain knowledge well, but the task specificity is not taken into account in the finetune stage, and the model is difficult to do a good response matching. For this reason, we propose UOS, which integrates the understanding of utterance in both post training and finetune stages. Our contributions are summarized as follows.

- We consider the specificity of multi-turn response selection and recognize the problem that pre-training models in the finetune stage have insufficient understanding of utterance and cannot do a good job of answering matching. For this reason, we propose UOS that integrates the understanding of utterance in both post training and finetune stages.

- We not only verify the accuracy of UOS over baselines on NOESIS II task, but also we prove the stable performance of UOS by Chinese datasets E-commerce and Douban.

2. Related Work

Building an automated conversational agent is a long-standing goal of artificial intelligence. Previous research includes task-oriented conversational systems that focus on completing tasks in vertical domains; and chatbots that aim to have consistent and natural conversations with humans on topics in open domains. Existing dialogue system models can be classified into two main categories: retrieval-based [12] and generation-based [14]. Retrieval-based systems have the advantage of being informative and responsive, as they select the next most appropriate response possible in the set of candidate responses. Generation-based approaches synthesize a response with a natural language generation model by maximizing its generation probability, given the previous conversational context. Most of the recent studies have focused on response selection for multi-turn conversations, where the main task is to select the most appropriate response from the candidate set given the conversation context. Zhou et al [27] provide a straightforward single-turn-to-multi-turn idea : combining multiple rounds of question and answer statements into one column, separated by SOS at the join, and treating the entire conversation history as " one utterance" to match the next utterance. In [23], a sequence matching network (SMN) is proposed, which first matches responses to each discourse and then accumulates matching information through a recurrent neural network. Unlike previous studies, the DUA model [26] distinguishes the importance of previous conversations for the first time and accumulates a large number of parts from each discourse based on each word of the discourse itself in a multi-turn scenario.

The Dialogue System Technology Challenge (DSTC) was initiated by scientists from Microsoft and Carnegie Mellon University in 2013 to drive the improvement of dialogue technology in academia and industry, and has great authority and popularity in the dialogue field. The proposed method in this paper is based on Track 2 (NOESIS II) of the DSTC8 competition [9]. The main task of this track is response selection: construct a

dialogue model that, given the dialogue content, selects from a candidate set the response most likely to appear in the next utterance. SA-BERT [5] is proposed in order to make the model aware of the speaker change information, which is an important and intrinsic property of multi-turn dialogues. [7] propose a new fine-grained post-training method that learns the utterance level interactions by training every short context-response pair in a dialogue session. ResSel [21] focus on the post-training stage, which can helps the model to train contextualized representations on domain-specific corpus and words that do not appear in general corpus.

It can be seen from this, pre-trained models have been used with great success for dialogue system response selection problems. One of the most representative is BERT [19], a bi-directional Transformers-based encoder with multiple layers. In this paper we use the BERT model, which is designed to be easily applied to other nlp tasks by way of fine-tuning. Since it is pre-trained on a large general text corpus (e.g. Wikipedia corpus), the general language representation is learned. Therefore sufficient supervision of task-specific words and phrases during fine-tuning is not enough. To alleviate this problem, we perform post- training BERT on a task-specific corpus, which helps the model understand the specific domain. In this paper, we draw on other papers to train with three objectives, MLM, NSP, and URC, which can effectively learn contextual representations from the corpus. Among them, MLM is trained by randomly masking off tokens from parts of the corpus, thus allowing the model to learn the overall contextual representation; the NSP approach allows the model to learn two text sequences A and B, whether the latter is the next sequence of the former, and the predicted result [CLS] tokens the output. URC learns each (context,response) pair with utterance-level interaction information.

3. Utterance-oriented Strategy

We propose an Utterance-oriented Strategy (UOS) to solve the multi-turn dialogue response selection problem. The UOS consists of two stages including post-traing and fine-tuning. First, we post-train the model so that it could learn more about the domain information and be more adaptable to the fine-tuning tasks. Then, during the fine-tuning process, we encode the dialogue and response into Bert to get a token-level dialogue and response representation, then we get the utterance-level dialogue and response representations through the utterance-aware matching network, finally, we have an utterance-aware score for each response.

Our UOS is better than the existing approaches in the following ways. First, we use a more granular post-training, which can better capture the coherence and semantic information between utterances in the context. Secondly, many current studies of multi-turn dialogue response selection based on the pre-training models are directly using the [CLS] tag output in the fine-tuning part, which only focus on the token-level interactions and never pay attention to the interaction between utterances. In this paper, we design a multi-channel matching network based on utterance level, which makes the model making full use of the information of the tokens and utterances. By combining this utterance-oriented post-training and fine-tuning strategy, our approach better captures the relationship between the context and response.

3.1. Problem Formulation

Suppose that the dataset $D = \{(c, r, y)_i\}_{i=1}^{N}$is a set of N triples that consist of the context c_i, candidate response r_i and the ground truth label y_i. The context $c_i = \{u_1, u_2, \ldots, u_M\}$ consists of a sequence of utterances, M represents the maximum context length. The utterance $u_k = \{w_{k,1}, w_{k,2}, \ldots, w_{k,L}\}$consists of a sequence of word tokens, L represents the maximum utterance length. Candidate response $r_i = \{w_1, w_2, \ldots, w_n\}$ is a single utterance, n represents the number of word tokens in the candidate response. Ground truth label $y_i \in \{0, 1\}$, if the candidate response r_i is the correct response to the context c_i, $y_i = 1$, otherwise $y_i = 0$. The multi-turn dialogue response selection task is to give a matching score between the candidate response and the context, and then sort all the responses based on the score.

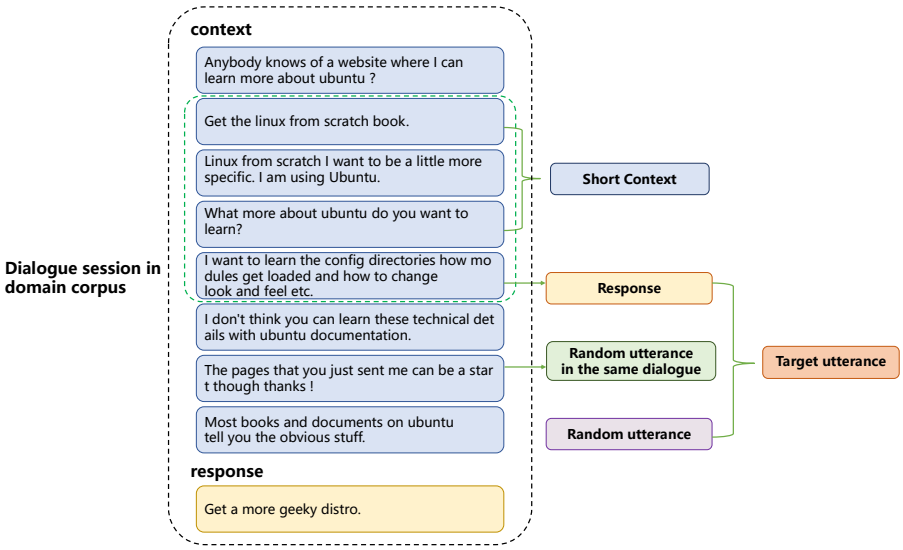

Figure 2. An URC example of post-training from the Ubuntu corpus.

3.2. Utterance-aware Post-Training

In this section, we introduce the utterance-aware post-training of UOS. The NSP training target can effectively learn the relationship between two sequences, if it is used in the multi-turn dialogue response selection task, the method is to separate the context and the candidate response, so that the model can learn the relationship between the context and response. However, this kind of training method is coarse-grained, because the context is composed of many turns of utterances, the model can't learn the coherence of the context and response at the utterance level. In addition, the model can't learn the semantic relationship between the utterances in the context well. Inspired by [7], we replace the NSP with URC, which allows the model to better understand the semantic meaning and coherence of context in a more granular way at the utterance level.

In our work, we use both MLM and URC in post-training which can better learn the related domain information and more adaptable to the fine-tuning task. About MLM, Fig. 3 gives an example of use MLM in post-training from the Ubuntu corpus. MLM randomly Mask the token in the corpus, so that the model can learn more domain-related contextual representations.

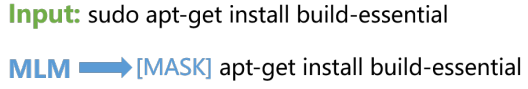

Figure 3. A MLM example of post-training from the Ubuntu corpus.

About URC, we follow the training method in [7], the corpus which consists of M utterances are divided into several short context and response pairs for training.We denote short context and response pairs $(\mathrm{SC}, \mathrm{R})_j = \{(u_j, u_{j+1}, \ldots, u_{j+k-1}), u_{j+k}\}$, URC classified the target utterance u_t into three labels. The first label is a random utterance selected from the corpus u_r. The second label is a random utterance selected from the same context except the correct response $u_s, s \in [1, M]$ and $s \neq j + k$. The third label is the correct response u_{j+k}. Since there should be similar topics between the utterance and the correct response in the same context, URC learns the topic

information of the context by classifying the first label random utterances in the corpus and the third label correct response. Besides, URC learns the coherence of the context by classifying the second label random utterances and the third label correct response in the same context. Fig. 2 gives an example of use URC in post-training from the Ubuntu corpus.

We add [EOT] at the end of each utterance so that the model can capture the end information. Besides, we add [SEP] between the context and response so that the model can capture their differences.The input sequence can be expressed as:

$$\text{input } = [CLS]u_j[EOT]\ldots u_{j+k-1}[EOT] [SEP]u_t[SEP] \quad (1)$$

Then we output [CLS] to the linear layer to get a matching score $\text{score}\,(\text{sc}, u_t)$. The Loss function of URC can be described as:

$$L_{URC} = -\sum\sum_{i}^{3} y_i \log\left(\text{score}\,(\text{sc}, u_t)_i\right) \quad (2)$$

Finally, we add the loss of MLM and URC to get the final post-traing loss.

3.3. Utterance-aware Fine-Tuning

In this section, we introduce the utterance-aware fine-tuning of UOS. The overall fine-tuning approach is described in Fig. 4.

3.3.1 BERT-based Encoder

We use the post-training BERT as our encoder. Similar as the post-training, we use [EOT] and [SEP] as the separator. The input sequence can be expressed as:

$$\text{input } = [CLS]u_1[EOT]u_2[EOT]\ldots u_M [EOT][SEP]r_i[SEP] \quad (3)$$

At first, input is a sequence of words, then we encode input through the token embedding layer,segment embedding layer and position embedding layer. Finally, we add the encoding result of the three layers to get the final input vector. The encoders in BERT use the self-attention mechanism to calculate the relationship between the context and response at the token level.

3.3.2 Context-response Matching Network

We use the hidden vector output of the last layer in BERT as the context-response representation that aggregates the token-level concerns, it can be expressed as:

$$\text{output } = [CLS]h_{u_1}[EOT]h_{u_2}[EOT]\ldots h_{u_M} [EOT][SEP]h_{r_i}[SEP] \quad (4)$$

To further capture the interaction between context and response at the utterance-level, we first split the token-level output into utterances by using separators, then we get the utterance vector representation of the same dimension d by simple average pool aggregation.The utterance vector can be expressed as:

$$s_j = \text{MeanPool}\,(h_j) \in \mathbb{R}^d, j = u_1, u_2, \ldots, u_M, r_i \quad (5)$$

Then we divide the utterance vector representation into two parts h_c and h_r according to the separator.

$$h_c = s_j, j = u_1, u_2, \ldots, u_M \quad (6)$$

$$h_r = s_j, j = r_i \quad (7)$$

Because there are many long utterances in the context, we design a multi-head perceptive matching network to capture the interaction in the context at utterance-level. Multi-head perceptive matching networks are composed of linear layers of different dimensions. The specific description of the i-head is as follows:

$$g_i = \sigma\left(W_{\text{i2}}\delta\left(W_{\text{i1}}h_c\right)\right) \quad (8)$$

In order to further capture the interactive information between response and context, let the response focus on the more important utterances in the context, we design a utterance-level response-based attention.

$$\text{att } = \sigma\left(W_{\text{r}}h_r + b\right) \quad (9)$$

We add all utterance representations obtained by multi-head perceptive matching network to get a

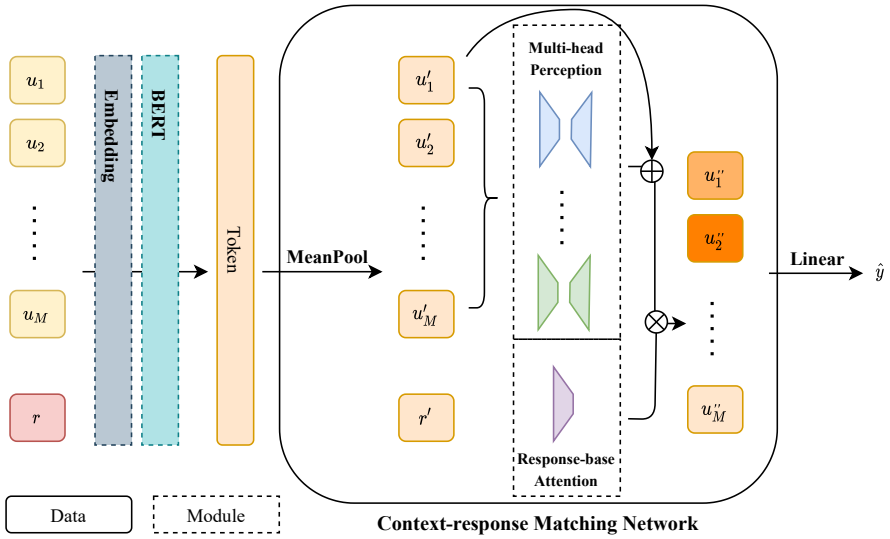

Figure 4. Fine-tuning BERT for Response Selection

more abundant utterance representation. Then we use the response-based attention to pay more attention to the more important utterance in the context. We represent the utterance-aware dialogue that eventually merges the attention of the response as d:

$$\mathrm{d} = \mathrm{att} * (\sum_{i=1}^{3} g_i + h_c) \tag{10}$$

Finally, we score the utterance-level context vector which incorporates the attention of response through the linear layer to select the correct response.

4. Experiments

4.1. Datasets and Evaluation Metrics

We conducted experimental tests on four publicly available pairwise multi-turn response selection datasets, including Ubuntu Corpus V1Douban Conversation Corpuse-commerce Dialogue Corpus and DSTC8-Track2-Subtask1 Ubuntu corpus. Dataset-specific information is provided in Table 1

- Ubuntu Corpus V1 is a dataset containing a large number of text-based conversations between humans and AI agents. These conversations are designed to simulate natural conversations between humans and AI agents in a variety of different contexts, including customer service, technical support, and general conversation.

- Douban Corpus is a collection of data related to the Douban website, a popular social networking platform in China. It may contain information about users and their interactions on the site, such as posts, comments, and ratings.

- E-commerce Corpus is a collection of data related to online shopping. It may contain information about products, prices and transactions on an e-commerce platform.

- DSTC8 Ubuntu is a multi-party conversation dataset of users discussing technical issues with the Ubuntu operating system, such as ubuntu hardware support, nvidia installation, certificates and other issues.

Table 1. Statistics of the datasets for the experiments

Dataset	Train	Valid	Test	Response options
Ubutnu V1	1M	0.5M	0.5M	10
Douban	1M	50k	10k	10
E-commerce	1M	10k	10k	10
DSTC8 Ubuntu	22M	0.5M	0.5M	100

The model performance is mainly measured by the average of recall rates Recall and mean reciprocal rank MRR . The recall Recall@K is the ratio of the number of relevant results retrieved from the topK results to the number of all relevant results in the library, which indicates the system's search completion rate. And the MRR index can be derived from the following equation:

$$MRR = \frac{1}{Q}\sum_{i=1}^{|Q|}\frac{1}{rank_i} \tag{11}$$

where Q is the total number of questions in the evaluation set, $rank_i$ indicates the position of the first correct answer predicted for the ith question Q_i in the answer set, and 0 if there is no correct answer for question Q_i, which means the more forward the position of the correct answer, the better the performance is.

4.2. Experimental Results

We test the proposed model UOS and the previous model on the Ubuntu Corpus V1 dataset, Douban dataset and e-commerce dataset respectively. The specific evaluation results are shown in Table2 and Table3. We reproduce the results of BERT, BERT-DPT and URC to make a fair comparison as far as possible. It should be noted that other model results are from the existing literature due to time and resource constraints. As for the implementation details, The batch size is set to 32 and the maximum sequence length is set to 512, specifically 448 for a dialog context and 64 for a response. In addition, the Adam weight decay optimizer is used to optimize the model, with a learning rate of 3e-5. During the Responce Selection fine-tuning process, the maximum number of utterances in a conversation is set to 10.

Table 2. Evaluation results of UOS and previous methods on the Ubuntu Corpus V1

Dataset	Ubuntu Corpus V1			
Metrics	**R_{10}@1**	**R_{10}@2**	**R_{10}@5**	**MRR**
SMN	0.726	0.847	0.961	-
DUA	0.752	0.868	0.962	-
DAM	0.767	0.874	0.969	-
MRFN	0.786	0.886	0.976	-
IoI	0.796	0.894	0.974	-
MSN	0.800	0.899	0.978	-
BERT-Base	0.809	0.899	0.975	0.879
BERT-DPT	0.848	0.926	0.984	0.906
SA-BERT	0.855	0.928	0.983	-
URC	0.877	0.945	0.989	0.925
UOS	**0.883**	**0.947**	**0.991**	**0.929**

As shown in Table2 and Table3, the evaluation results of our proposed model show that it outperforms exsiting state-of-the-art results in all metrics for Ubuntu Corpus V1 as well as Douban Corpus and E-commerce Corpus. Specifically, it attains performance improvement by 1.4% , 0.2%, 0.2% in terms of R_{10}@1, R_{10}@1 and R_{10}@5 respectively in Ubuntu corpus, compared to the previous URC method. Meanwhile, the proposed UOS strategy surpass the present methods by large margins of 2.7% R_{10}@1 and 2.7% MRR on Douban Conversation Corpus, 2.8% R_{10}@1 and 1.6% MRR on E-commerce Corpus. This is because our proposed model further considers the relationship between utterances in the process of fine-tuning to better understand the dialogue. These results shows that the proposed model has better performance on dialogue response selection tasks and has good adaptability to different corpus datasets.

4.3. Ablation Study

4.3.1 Post Training

In this section, we compare the effectiveness of different post-train process on Ubuntu Corpus V1. The results are shown in Table4. It's observed that the post-train procedure we apply in UOS model is the most effective. In general, MLM is to cover up some Tokens randomly and let the model predict so that it can learn the context representation while

Table 3. Evaluation results of UOS and previous methods on the Douban Corpus and E-commerce Corpus

Dataset	**Douban Corpus**				**E-commerce Corpus**			
Metrics	$\mathbf{R}_{10}$@1	$\mathbf{R}_{10}$@2	$\mathbf{R}_{10}$@5	**MRR**	$\mathbf{R}_{10}$@1	$\mathbf{R}_{10}$@2	$\mathbf{R}_{10}$@5	**MRR**
BERT-Base	0.233	0.417	0.736	0.541	0.495	0.684	0.914	0.666
BERT-DPT	0.245	0.430	0.757	0.557	0.539	0.734	0.951	0.702
URC	0.230	0.426	0.755	0.544	0.761	0.907	0.985	0.858
UOS	**0.257**	**0.444**	**0.767**	**0.571**	**0.789**	**0.912**	**0.985**	**0.874**

NSP is created to the relationship between two utterances. Therefore, they can help model predict the response to a certain extent. However, the post-train applied in the proposed UOS model not only contain the MLM and NSP tasks, it also includes subject and consistency prediction, which leads to an excellent post-train model outcome.

Table 4. Evaluation results of different post-train methods on Ubuntu Corpus V1

Dataset	**Ubuntu Corpus V1**			
Metrics	$\mathbf{R}_{10}$@1	$\mathbf{R}_{10}$@2	$\mathbf{R}_{10}$@5	**MRR**
Base	0.809	0.897	0.975	0.879
NSP	0.815	0.902	0.976	0.883
MLM	0.819	0.905	0.978	0.886
MLM+NSP	0.848	0.926	0.984	0.906
MLM+URC(ours)	**0.877**	**0.945**	**0.989**	**0.925**

4.3.2 Robustness Test for Larger Response Candidate Pool

In order to verify the robustness of the proposed model and test its response selection ability in a larger candidate pool, we select the Ubuntu dataset in DSTC8-Track2-Subtask1 to further verify the model performance. In specific, we compare the proposed UOS model performance wIth previous method on DSTC8-Track2-Subtask1 Ubuntu dataset. Final results are shown in Table5.

Table 5. Evaluation results on DSTC8 Ubuntu Dataset

Dataset	**DSTC8 Ubuntu**			
Metrics	$\mathbf{R}_{100}$@1	$\mathbf{R}_{100}$@2	$\mathbf{R}_{100}$@5	**MRR**
DSTC Baseline	0.212	0.290	0.421	0.323
BERT	0.258	-	-	0.335
SA-BERT	0.477	-	-	0.594
UOS	**0.548**	**0.650**	**0.769**	**0.648**

According to the evaluation results on DSTC8-Track2 Ubuntu dataset, the proposed UOS model performs better than other methods when the number of response options in candidate pool expands to 100. It achieves 0.548 R_{100}@1 and 0.648 MRR which has a significant enhancement compared to SA-bert approach. Consequently, it can be proved that our proposed UOS model has better robustness for response-selection task.

5. Conclusion

In this paper, we have applied the fine-grained post-training method in the post-training and fine-tuning stage. Furthermore, we propose UOS, getting the interactive relationship between context and response at the utterance level. The proposed method allows the model to learn the semantic relevance of the utterances between context and response in the dialogue, and it improves the model's performance in response-selection task. The experimental results on the Ubuntu V1 dataset show our model's superiority. In addition, the outcome of the model on the Douban and E-commerce Corpuses demonstrates its generality and stability.

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